

6 The curve  $C$  has polar equation  $r^2 = \tan^{-1}\left(\frac{1}{2}\theta\right)$ , where  $0 \leq \theta \leq 2$ .

(a) Sketch  $C$  and state, in exact form, the greatest distance of a point on  $C$  from the pole. [3]

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(b) Find the exact value of the area of the region bounded by  $C$  and the half-line  $\theta = 2$ . [5]

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(c) Show that, at the point furthest from the half-line  $\theta = \frac{1}{2}\pi$ ,

and verify that this equation has a root between 0.6 and 0.7. [5]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.